Sticks and Stones

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Field(s) of Interest: Structural Engineering, Civil Engineering

Brief Overview:
This lesson introduces students to the basics of structural engineering by looking at how bridges and buildings safely support loads and withstand natural disasters. By the end of this lesson, mentees should understand how the shape and design of a structure affect how stable it is.

Agenda:
- Introduction (5 min)
- Module 1: Truss me this bridge works (10 min)
- Module 2: Al Dente Engineering (25 min)
- Module 3: Oh No Earthquake :| (15 min)
- Conclusion (5 min)

Main Teaching Goals/Key Terms:

- ➔ Force
- ➔ Truss
- ➔ Load Distribution
- ➔ Structural Engineering
- ➔ Beam
- ➔ Joints
- ➔ Center of Gravity
- ➔ Factor of Safety
- ➔ Load Path
- ➔ Earthquake
- ➔ Engineering Design Process
- ➔ Stability

Background for Mentors

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Module 1

- Force
- Truss
- Load Distribution
- Structural Engineering

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Force is a push or pull on an object. In this lesson, the main force we will be looking at is gravity, which constantly pulls objects towards the Earth.

$$F = m\,a$$

In structural engineering, gravity is the force/load on the structures.

A **truss** is a structural framework of beams arranged into a pattern of triangles. Unlike squares or rectangles, triangles do not change shape when a force is applied. When a force is applied, the load is distributed along the sides of the triangle rather than being concentrated at a single point.

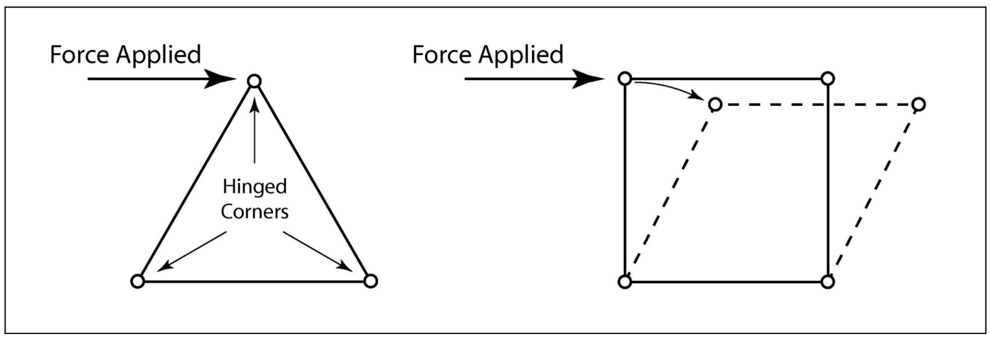

Figure 1: A visual depiction of how a triangle and a square would react to a force.

Load distribution refers to spreading a weight across multiple points to prevent a failure in the system. In the case of module 1, using the accordion fold allows the load to spread, making the structure stronger.

Structural Engineering is a specialized branch of civil engineering that focuses mainly on designing structures that can safely support loads and withstand forces.

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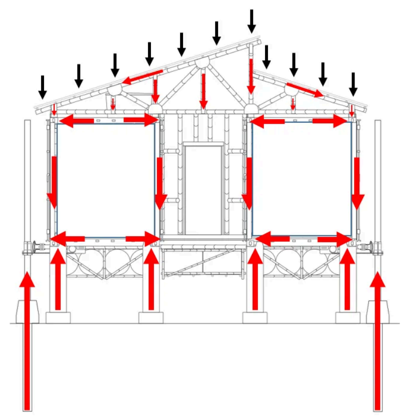
Module 2 ● Beam ● Joints ● Center of Gravity ● Load Path	A **beam** is a structural element designed to support a load while **joints** are the points where the beams connect. The joints allow forces to travel through the structure. The **center of gravity** is a point on an object where the total weight is assumed to be concentrated at. Structures with a wider and lower center of gravity tend to be more stable and reduce the chances of the structure tipping over. **Load path** is the continuous route a force takes through a structure to the ground. When a load path is broken (e.g. weak joint) the building fails and could potentially collapse.  **Figure 1:** An example of a load path in a structure

Module 3

- Earthquake
- Stability
- Engineering Design Process
- Factor of Safety

An **earthquake** occurs when energy stored in the Earth's crust is released due to movement along tectonic plates. This movement causes the ground to shake. It is the goal of the structural engineer to ensure a building can withstand an earthquake. This is why the **stability** of a structure is so important as it needs to remain upright and resist any sort of tipping, sliding, etc.

A structure's stability can be influenced by having a wider base, lower center of gravity, strong joints, and more!

During this module, the mentees will be able to follow the **Engineering Design Process** which highlights testing their spaghetti builds and improving upon the design.

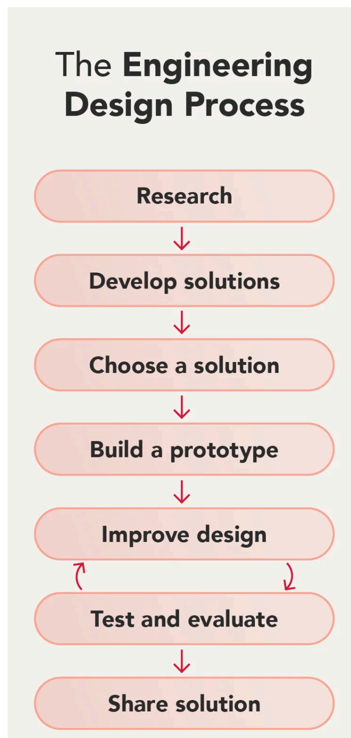


Figure 1: The steps of the EDP

The **factor of safety** is a metric that tells us how much stronger a system is than it needs to be for its intended load. It is modelled using this equation:

$$FoS = \frac{Structural\ Capacity}{Actual\ Load}$$

Engineers are expected to always design buildings to be stronger than they need to be in order to account for unexpected events.

Introduction

This lesson is important because structural engineering is all around us as we drive across a bridge or enter a building, it is important to stop and think about how and why they work. This infrastructure needs to be designed to hold weight and withstand natural forces.

Concepts to Introduce ● Load Distribution ○ Why can someone lay on a bed of nails and not get hurt? ■ Your weight is distributed across all the nails so it doesn't hurt as much ● Truss ○ Refer to methods for teaching in module 1 for the human demonstration	**Questions to Pique Interest** ● How are skyscrapers able to be so tall and not fall? ● What shapes do you think are the strongest? ● What might happen to buildings during an earthquake?
Scientists, Current and Past Events ● Key inventors such as William Howe Thomas and Caleb Pratt, James Warren have all contributed to the development of the truss. ○ History of a truss bridge ● The Empire State Building was designed by Homer G. Balcom designed the building's steel frame to handle significant lateral wind loads for extended periods of time. ○ Read More Here	**Careers and Applications** ● Structural Engineer ○ Designs bridges, buildings, and other structures to safely support loads ● Civil Engineer ○ Works on infrastructure projects like roads, bridges, dams, etc ● Architect ○ Designs the appearance and layout of buildings ● Seismologist ○ Studies earthquakes and how seismic waves move throughout Earth

Module 1: Truss me this bridge works

Each mentee is given a singular sheet of paper and the goal is to have the paper span a gap and hold a weight. This is to show how trusses are used to create rigid structures.

Teaching Goals

1. **Force:** A push or pull on an object
2. **Truss**: A structural framework of beams arranged into a pattern of triangles to create a rigid unit
3. **Load Distribution:** The spread of forces across multiples points in a structure
4. **Structural Engineering:** The study of making structures safe and stable

Materials

- 1 sheet of paper per mentee
- 1 wooden block per mentee
- 1 roll of tape per site

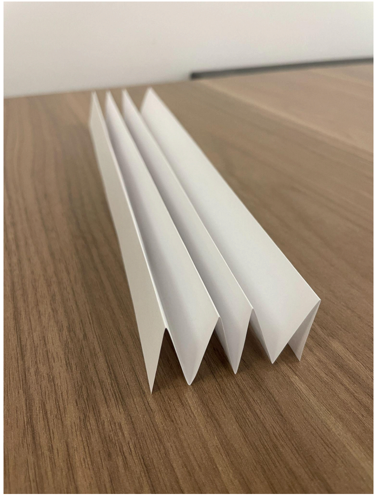
Different Methods for Teaching

1. **Setting the Pace:** If your site is worried about running out of time, you can adapt this activity to a classroom demo.
2. **Breaking it down:** To demonstrate how triangles help with load distribution, have mentors demonstrate a human square (feet together) or human triangle (feet wider than shoulder width apart). Another mentor will gently push down on the shoulders and also sideways to demonstrate how having the legs spread helps to distribute the force along the lengths of the leg.
    a. This demonstrates that the person/structure is strong against purely downward forces but it is weak against sideways forces.
3. **Analogy**: When a pizza is folded into a taco shape, it creates a V (partial triangle), allowing it to stay rigid.

Procedure

1. Mentees are given a singular sheet of paper and the goal is to have the paper span a gap and hold a weight (singular wooden block)
    a. If desks cannot be moved, either set up two stacks of books or chairs with a 6-7 inch gap
2. Have students lay the flat sheet of paper across. Then try to place a wooden block on it (it will fail)
3. Finally make an accordion fold (series of **triangles**)
    a. Fold the paper in half hotdog style,
    b. Fold it hotdog style in half again, then

Figure 1: *"The accordion fold (triangular pattern)*

once more. c. Unfold the paper and then refold along the partitioning accordion style 4. Your site can choose to do a demo as well to show why a square is not as strong as a triangle (see Figure 3) a. Fold a paper into a 4ths and tape the edges together. b. Place a weight on top and watch it deform	*easily holds a load"*  **Figure 2:** *"How your paper should look like"*  **Figure 3:** *"A square fold easily deforms when a load is added"*

Classroom Notes

For the accordion fold, make sure that there are more points of contact with the desk.

Module 2: Al Dente Engineering

In this group build, mentees will use their knowledge on trusses and [content from this module] to build a two story structure that will eventually be able to withstand the "earthquake" in module 3.

Teaching Goals	Materials
1. **Beam**: A structural element designed to carry the load 2. **Joints:** The marshmallows hold the beams together 3. **Center of Gravity:** A point on an object where the weight is assumed to be concentrated at 4. **Load Path:** The route the forces take as they move through a structure to the ground	• 12 pieces of spaghetti per group • 4 Large Marshmallows per group • 12 mini marshmallows per group

Different Methods for Teaching

1. **Troubleshooting:** If a tower is learning, ask the mentees: Where is most of the weight? Is the bottom wider or thinner than the top?
2. **Ensuring fair participation:** It is not too uncommon for one or two mentees to take charge of the group while others don't get to participate. If you think this will be a problem at your site, you can assign roles and have them rotate every 2-3 minutes if possible. Have one mentor at each group to facilitate this.
    - Potential roles (pick some): 1-2 architects (help design the tower), 1-2 Builders, material manager (break spaghetti, rip marshmallows etc.), 1 timekeeper, 1 shake tester (points out where structure is unstable)

Procedure

1. Split the mentees into groups of 4-5 students
2. Hand out 4 large marshmallows, 12 pieces of spaghetti, and 12 mini marshmallows to each group
3. Mentees will use the given materials to build a two story structure that they will test in module 3 (let them know ahead of time that this is the goal of their build)
    a. Remind them of the truss from Module 1 to help with the structural integrity of their builds
4. Have the mentees use the larger marshmallows at the bottom of the build, effectively lowering the center of gravity
5. See Figure 1 and Figure 2 for an example of sample builds

Figure 1: *“Example Build with 4 points of contact”*

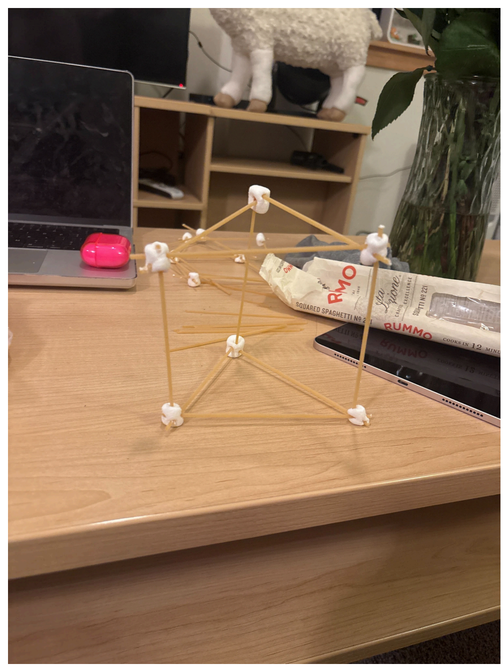
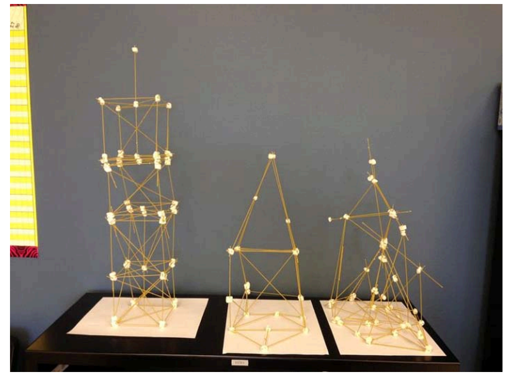


Figure 2:”Example build with 3 points of contact"

	 **Figure 3:** *"Image found online of sample builds"*

Classroom Notes

The spaghetti can easily break if it is too long so let the mentees know it is easier to work with that way, or break the spaghetti in half before giving them to the mentees.

Module 3: Uh Oh Earthquake :|

In this group activity, students will test the structures they made in Module 2 to see if it can withstand a simulated earthquake.

Teaching Goals	Materials
1. **Earthquake:** The ground shaking due to tectonic plates in the Earth's crust sliding or colliding 2. **Stability:** A structure's ability to remain upright 3. **Engineering Design Process:** A step-by-step process engineers follow to improve designs 4. **Factor of Safety:** How strong the building needs to be in order to stand up and withstand the earthquake	• 4 golf balls per site • 3 rubber bands per site • 3 sheets of cardboard per site

Different Methods for Teaching

1. **Stepping down the activity:** For sites that struggle with the build, you can instead have mentees look to modify their build to be more stable instead of doing the shake test. This ties it back to the teaching goals and could avoid mentee frustration.
2. **Shake it up:** If your site is advanced, feel free to bring additional information to site. For mentees curious to know more: When designing buildings and infrastructure, engineers must consider how the structure handles varying forces, including from wind and earthquakes.

Procedure

1. Mentors will insert 4 golf balls between the two sheets of cardboard to create the earthquake table
2. Place the student's tower on the earthquake tester and shake lightly until the structure either falls or breaks
3. Have the mentees use the Engineering Design Process to revise their structure and then retest them

Figure 1: *"This is how the earthquake table should look like"*

	 **Figure 2:** *"The golf balls between the cardboard"*

Classroom Notes

The golf balls can move around easily so try not to turn the cardboard vertically. It can be inserted by placing it on the table and lifting the top sheet and inserting the golf balls.

Conclusion

Ask mentees which structure shape held the most weight? What made certain towers more stable than others? Reinforce the idea that engineers need to test and improve designs many times to make structures strong.

Summary Materials Table

Material	Amount per Site	Expected $$	Vendor (or online link)
Spaghetti	12 per group	$10	Amazon
Large Marshmallows	4 per group	$50	Amazon
Golf balls	4 per site	$32	Amazon
Mini Marshmallows	12 per group	$7	Amazon
Paper	1 per mentee	$50	Amazon
Cardboard	3 per site	$22	Amazon
Tape	1 roll per site	$0	N/A
Wooden Blocks	1 per mentee	$0	N/A